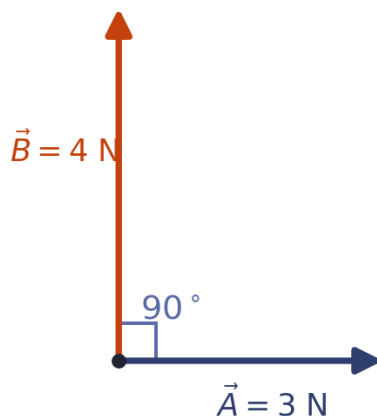


Vector Addition — Find the Resultant

Geometric method · no coordinate system · Class 11 — JEE Foundation

For each diagram, two vectors **A** and **B** act from a common point with the angle between them marked. Find the magnitude (and direction) of the resultant using

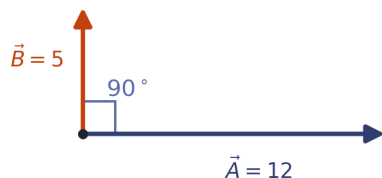
$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$ and $\tan \alpha = (B \sin \theta) / (A + B \cos \theta)$, where α is the angle of **R** from **A**. Take $\sqrt{2} \approx 1.41$, $\sqrt{3} \approx 1.73$.



Question 1

Two forces **A** and **B** act at a point with $|A| = 3 \text{ N}$ and $|B| = 4 \text{ N}$, inclined at 90° to each other (as shown).

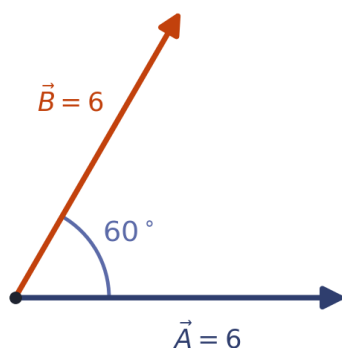
Find: the magnitude and direction of the resultant force **R**. (Direction = angle measured from **A**.)



Question 2

Two vectors **A** and **B** with $|A| = 12$ units and $|B| = 5$ units are inclined at 90° (as shown).

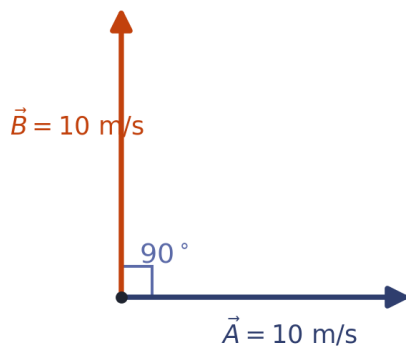
Find: the magnitude and direction of the resultant **R**. (Direction = angle measured from **A**.)



Question 3

Two vectors of equal magnitude, $|A| = |B| = 6$ units, are inclined at 60° (as shown).

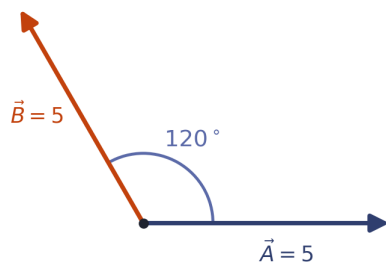
Find: the magnitude and direction of the resultant **R**. (Direction = angle measured from **A**.)



Question 4

Two velocities with $|\mathbf{A}| = |\mathbf{B}| = 10 \text{ m/s}$ act at 90° to each other (as shown).

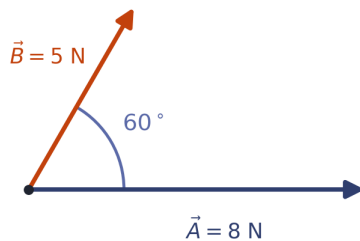
Find: the magnitude and direction of the resultant velocity $\mathbf{R}$. (Direction = angle measured from $\mathbf{A}$.)



Question 5

Two vectors of equal magnitude, $|\mathbf{A}| = |\mathbf{B}| = 5$ units, are inclined at 120° (as shown).

Find: the magnitude and direction of the resultant $\mathbf{R}$. (Direction = angle measured from $\mathbf{A}$.)



Question 6

Two forces with $|\mathbf{A}| = 8 \text{ N}$ and $|\mathbf{B}| = 5 \text{ N}$ act at 60° to each other (as shown).

Find: the magnitude and direction of the resultant force $\mathbf{R}$. (Direction = angle measured from $\mathbf{A}$.)

Solutions

$\cos 90^\circ = 0$, $\cos 60^\circ = \frac{1}{2}$, $\cos 120^\circ = -\frac{1}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2} \approx 0.866$. Final answers in **bold**.

1. $R = \sqrt{3^2 + 4^2 + 2 \cdot 3 \cdot 4 \cdot \cos 90^\circ} = \sqrt{9 + 16 + 0} = \sqrt{25} = \mathbf{5 \text{ N}}$.
Direction: $\tan \alpha = (4 \sin 90^\circ) / (3 + 4 \cos 90^\circ) = 4/3 \rightarrow \alpha = \tan^{-1}(4/3) \approx \mathbf{53.1^\circ \text{ from A}}$.
2. $R = \sqrt{12^2 + 5^2 + 0} = \sqrt{169} = \mathbf{13 \text{ units}}$.
Direction: $\tan \alpha = 5/12 \rightarrow \alpha \approx \mathbf{22.6^\circ \text{ from A}}$.
3. $R = \sqrt{6^2 + 6^2 + 2 \cdot 6 \cdot 6 \cdot \cos 60^\circ} = \sqrt{36 + 36 + 36} = \sqrt{108} = 6\sqrt{3} \approx \mathbf{10.39 \text{ units}}$.
Equal magnitudes $\Rightarrow$ the resultant bisects the angle $\rightarrow \alpha = \mathbf{30^\circ \text{ from each vector}}$.
4. $R = \sqrt{10^2 + 10^2 + 0} = \sqrt{200} = 10\sqrt{2} \approx \mathbf{14.14 \text{ m/s}}$.
Equal magnitudes $\Rightarrow$ resultant bisects $\rightarrow \alpha = \mathbf{45^\circ \text{ from each vector}}$.
5. $R = \sqrt{5^2 + 5^2 + 2 \cdot 5 \cdot 5 \cdot \cos 120^\circ} = \sqrt{25 + 25 - 25} = \sqrt{25} = \mathbf{5 \text{ units}}$.
Equal magnitudes $\Rightarrow$ resultant bisects $\rightarrow \alpha = \mathbf{60^\circ \text{ from each vector}}$.
6. $R = \sqrt{8^2 + 5^2 + 2 \cdot 8 \cdot 5 \cdot \cos 60^\circ} = \sqrt{64 + 25 + 40} = \sqrt{129} \approx \mathbf{11.36 \text{ N}}$.
Direction: $\tan \alpha = (5 \sin 60^\circ) / (8 + 5 \cos 60^\circ) = 4.33/10.5 \approx 0.412 \rightarrow \alpha \approx \mathbf{22.4^\circ \text{ from A}}$.

Tip: when the two vectors are equal in magnitude, the resultant always bisects the angle between them — so its direction is simply $\theta/2$ from either vector (used in Q3, Q4 and Q5).